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The Surprising Effectiveness of Deleting Weights in LLM Reasoning and Adaptation

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Can deleting <0.05% of weights unlock full post-training performance? Yes!

Bit-Mask Tuning (BMT) learns which pretrained weights to erase, matching **Full Parameter Tuning (FPT)** at scale. Our analysis explains why pure deletion can behave like dense weight updates.

Takeaways



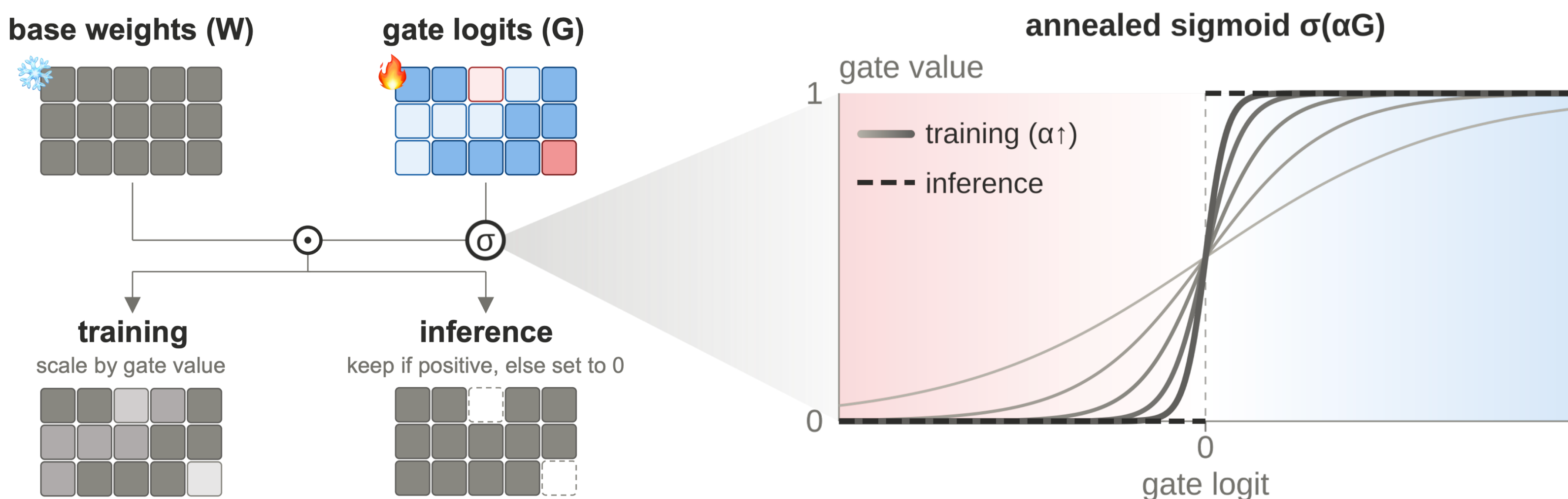
LLM Post-Training

Learned Sparsity

Forgetting

- Performance:** by learning what to delete, BMT matches full post-training at 7B/8B scale.
- Efficiency:** by storing only a compressed bitmask, BMT is 1000s of times smaller than FPT.
- Forgetting:** by making sparse deletions instead of dense updates, BMT preserves prior-task accuracy while FPT drifts.
- Understanding:** by inducing FPT-like update geometry, BMT tracks full fine-tuning more closely than LoRA or MTH.

Bit-Mask Tuning (BMT)



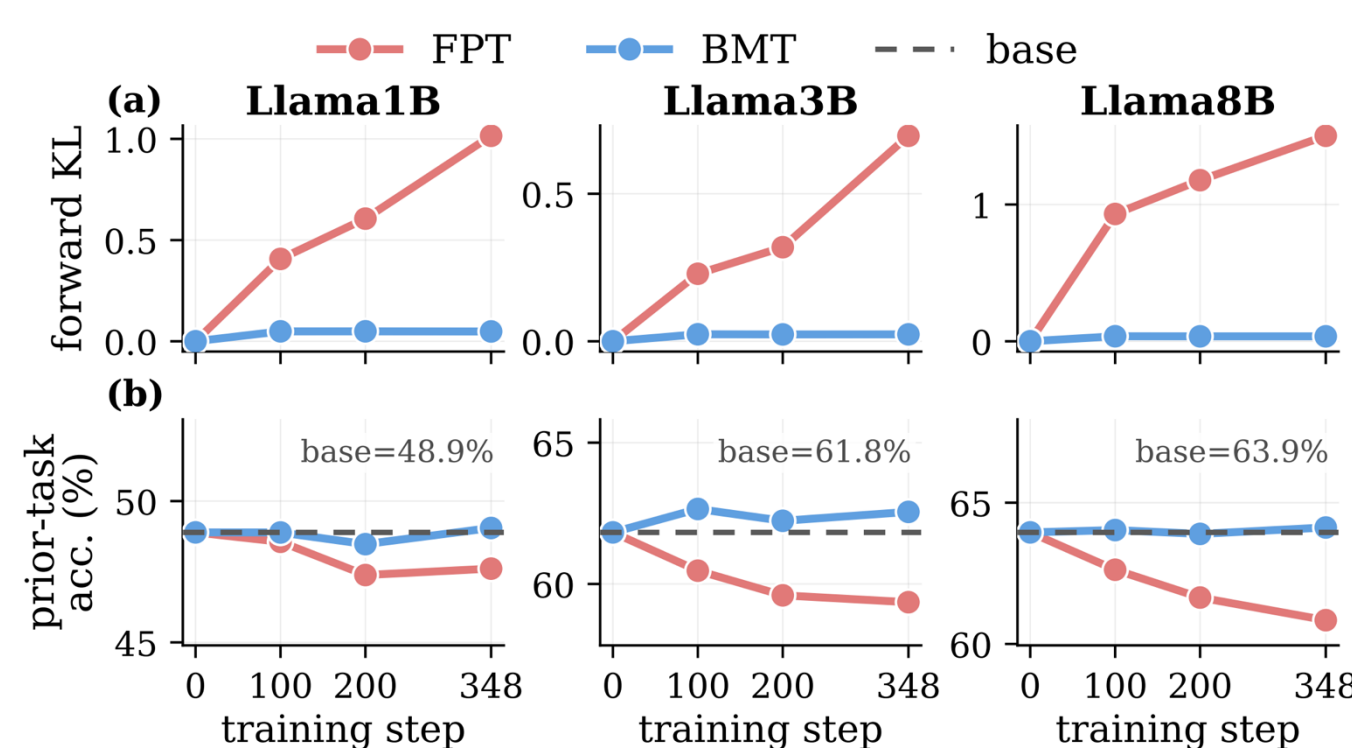
GRPO Post-Training Results

The performance gap between BMT and FPT vanishes as base model size grows.

Method	Llama, GSM8K						Qwen, SimpleRL-Zoo					
	1B		3B		8B		0.5B		3B		7B	
	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)
Base	1.21	0	53.53	0	60.27	0	10.44	0	37.73	0	40.18	0
FPT	60.50	4714	82.49	12256	83.24	30633	19.01	1885	44.35	11772	51.89	29051
LoRA r=1	59.29	2.7	82.39	5.8	83.38	10.0	20.30	2.1	43.55	7.1	52.13	9.6
LoRA r=4	60.58	10.8	82.44	23.2	82.87	40.0	20.30	8.4	43.59	28.6	51.70	38.5
MTH 99%	60.73	68.2	80.53	177.4	83.31	443.6	21.84	27.3	44.16	170.4	52.03	420.7
MTH 99.95%	58.61	4.1	82.17	10.7	83.07	26.6	20.16	1.7	43.87	13.7	51.24	25.3
BMT	54.44	1.0	81.49	3.1	83.55	8.4	17.64	0.5	41.98	2.7	52.32	6.9

BMT Forgets Less than FPT

FPT forgets from GRPO training; BMT does not.



Supervised Fine-Tuning and Self-Distillation Fine-Tuning

Beyond GRPO, BMT matches or exceeds FPT on a variety of SFT and SDFT settings; BMT transfers across post-training regimes.

Method	MMLU-Pro				BBEH			
	1B		3B		1B		3B	
	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)
Base	10.37	0	12.63	0	2.17	0	8.87	0
FPT	16.97	4714	18.35	12256	8.35	4714	11.39	12256
BMT	18.62	1.0	22.28	3.1	5.65	1.0	11.39	3.1

Method	ToolAlpaca		Science	
	Acc. (%)	Size (MB)	Acc. (%)	Size (MB)
Base	41.24	0	32.35	0
FPT	68.04	29051	71.20	29051
BMT	69.07	27.1	70.61	47.1

BMT Tracks FPT Geometry

Although BMT only deletes weights, its effective updates are closer to FPT than LoRA or MTH across **effective rank**, **spectrum drift**, and **principal-weight overlap**.

